

Aman Yadav

Software Development Engineer | Full-Stack Engineer | Golang | Next.js | Data Systems
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SUMMARY

Software Development Engineer with hands-on experience building a data analytics and workflow platform from the ground up. Progressed from frontend development into full-stack and backend engineering, working across Golang services, Next.js applications, data ingestion, workflow execution, real-time processing, and infrastructure. Experienced with Kafka, Redis, PostgreSQL, MongoDB, DuckDB, Docker, WebSockets, asynchronous processing, RBAC, subscriptions, schema evolution, idempotency, and external-system integrations.

EXPERIENCE

TuringX — Software Engineer

Bangalore / Remote | Dec 2025 – Present

Nexus — Data Analytics, Workflow & Real-Time Data Platform

- Built and extended **Nexus from the ground up**, progressing from frontend development into full-stack and backend engineering across Go services, data processing, workflow execution, and infrastructure.
- Developed and maintained **Golang APIs and backend services**, taking ownership of business logic, data access, asynchronous processing, integrations, and debugging across the platform.
- Designed and implemented a **DAG-based workflow execution engine** supporting data inputs, transformations, branching, API actions, and dataset/file outputs.
- Built application-level graph traversal and dependency resolution to compile workflow nodes into **SQL/CTE-based queries**, with DuckDB handling query execution across data sources.
- Implemented selective **materialization at workflow fan-out points** to avoid repeatedly evaluating expensive upstream transformations for downstream branches.
- Built transformation nodes for filtering, column selection/removal, limiting, derived columns, and schema transformations; implemented API outputs supporting single, per-row, and batch execution.
- Built ingestion flows for APIs, databases, webhooks, MQTT/IoT sources, and streaming systems, supporting manual, scheduled, and continuous synchronization.
- Worked with **Kafka-based event pipelines** for ingestion, validation, persistence, and real-time processing, including event idempotency using event IDs and composite/hash-based identities.
- Implemented schema evolution supporting additive fields, nullable fields, numeric type widening, and rejection of incompatible type changes while preserving previously valid events.
- Worked with **Redis-backed queues and workers** for asynchronous processing, including execution ownership, renewal, cancellation, retries, and server failover.
- Investigated failure scenarios involving **Kafka batching, offset commits, partitioning, event ordering, dataset-level failure isolation, retries, quarantine/DLQ handling, and duplicate processing**.
- Worked on the **Operational State Engine (OSE)**, investigating PostgreSQL snapshot/live Kafka race conditions, replay gaps, state regression, and duplicate aggregation during concurrent replay and live processing.
- Implemented **RBAC and multi-account access control**, including invitations, multi-account users, permissions, and account membership edge cases.
- Implemented SaaS subscription functionality using **Razorpay** and worked across PostgreSQL, MongoDB, Redis, Kafka, DuckDB, Docker, Linux, object storage, and CI/CD.

TuringX — Software Engineer Intern

Remote | May 2025 – Oct 2025

- Joined as a frontend-focused engineer working on the early Nexus dashboard and data visualization experience.
- Developed configurable charts, tables, KPIs, and analytical interfaces using **Next.js, React, JavaScript, Tailwind CSS, and reusable components**.
- Integrated frontend components with backend APIs and data-querying functionality while improving component organization, responsiveness, and maintainability.
- Built familiarity with the platform's backend and data model before transitioning into Golang and full-stack development.

- Built frontend websites and reusable components, including an art e-commerce website and blog platform with custom user-facing functionality.

- Built frontend components and website experiences for client projects, including the Bluesharktech brand agency website.

- Built responsive WordPress websites for clients, focusing on frontend implementation, cross-device compatibility, performance, and visual consistency.

PROJECT

- Built a cloud-based platform for building and deploying static web projects from Git repositories using Docker-based build environments.
- Developed the API and proxy services responsible for triggering builds, routing requests, and securely delivering generated build artifacts.
- Implemented real-time build log streaming using Kafka and WebSockets, allowing build output to be consumed while the build was running. Integrated object storage for generated artifacts and built the frontend dashboard using Next.js and Tailwind CSS.

TECHNICAL SKILLS

Languages: Golang, JavaScript, Python, SQL
Backend: Go, Gin, Node.js, Express.js, REST APIs, WebSockets, SSE
Frontend: Next.js, React.js, Tailwind CSS, Bootstrap, ShadCN UI
Databases: PostgreSQL, MongoDB, MySQL, Redis, DuckDB
Data & Systems: Kafka, event-driven processing, asynchronous processing, message queues, DAG/workflow execution, SQL/CTE generation, data ingestion, idempotency, schema evolution, real-time processing, DLQ/quarantine handling
Infrastructure: Docker, Git, CI/CD, Linux, SSH, DigitalOcean, object storage, service orchestration
Other: RBAC, SaaS subscriptions, Webhooks, MQTT, Razorpay, Figma, Webflow, WordPress

EDUCATION

Bachelor of Computer Applications (BCA)

Sahyog College of IT & Management,
Affiliated to KKSU University
Thane, Maharashtra, India

April 2026

RECOGNITION

- Site of the Day (SOTD), VICE Awards
- Site of the Month (SOTM), Web Design Awards
- Favorite Site / Nominee, Web Design Awards
- Nominee / Free Winner, Orpetron

Jan 2025 - April 2026